

Carbon at Risk

How CaR Makes Net-Zero Faster, Cheaper, and Safer:
An Academic Explainer of 10 Simulation Graphics and 10 Stakeholder Use-Cases

Working Paper — February 2026

Companion document to the CaR Net-Zero Interactive Simulation

The Carbon at Risk (CaR) framework provides a standardised, quantitative metric for assessing carbon dioxide removal (CDR) project risks—analogueous to Value at Risk (VaR) in financial markets. This document provides an academic-level explanation of ten simulation graphics comparing CaR-optimised versus non-optimised pathways to 2050 removal targets, followed by ten stakeholder use-cases with detailed VaR analogies. Each graphic isolates one MECE (mutually exclusive, collectively exhaustive) dimension of how risk quantification improves the net-zero trajectory across speed, cost-efficiency, and reliability.

Key Findings:

- **Faster:** CaR-optimised deployment reaches 8.5 Gt/yr by 2050 versus 5.2 Gt/yr without optimisation—a 3.3 Gt annual gap—by unlocking institutional capital 5–8 years earlier through standardised risk metrics.
- **Cheaper:** Dynamic, risk-based buffer pools and Wright’s Law cost acceleration reduce blended costs from \$95/t to \$45/t, saving approximately \$425 billion annually at scale.
- **Safer:** Portfolio diversification across 8 removal pathways with continuous CaR monitoring narrows the 95% confidence interval from $\pm 35\%$ to $\pm 12\%$, raising the probability of meeting the 7.5 Gt/yr target from 28% to 78%.

Sources: Gogerty & Johnson (2024) SSRN:4745542; Smith et al. (2024); Lee, Gogerty et al. *Nature Perspective*; Bauer & Rudebusch (2023) *Review of Economics & Statistics*; Groom & Venmans (2023) *Nature*; CaRGov Framework; CaR Policy Use Cases; CaRPS Proposal; *Forced Errors & Gains in Economic Potentials* (2024).

Contents

1. Introduction: The Case for Risk-Optimised Carbon Removal
2. Methodology: Simulation Framework and Assumptions
- Part A: Ten MECE Graphics — Optimised vs Non-Optimised**
3. G1: Removal Deployment Trajectory to 2050
4. G2: Blended Cost Per Tonne — Wright's Law Acceleration
5. G3: Probability of Meeting Interim Removal Targets
6. G4: Buffer Pool Efficiency — Static vs Dynamic
7. G5: Technology Portfolio Diversification
8. G6: Uncertainty Bands — Confidence Cone Narrowing
9. G7: Capital Mobilisation by Source
10. G8: Carbon Removal Insurance Market Creation
11. G9: Cumulative Warming Pathways
12. G10: Continuous Improvement Flywheel
- Part B: Ten MECE Stakeholders with VaR Analogs**
13. Stakeholder Analysis: CaR → VaR Mapping
14. Synthesis: The Compounding Effect
15. References

1. Introduction: The Case for Risk-Optimised Carbon Removal

Carbon dioxide removal (CDR) must scale from approximately 1.35 million tonnes per year today to 7–10 billion tonnes per year by 2050 to meet climate targets aligned with the Paris Agreement (Smith et al., 2024). This represents a roughly 7,000-fold increase in 25 years—an industrial scaling challenge comparable to building an entirely new sector nearly twice the size of global cement production (4.4 billion tonnes annually).

Current carbon removal evaluation frameworks rely on binary classifications—“permanent” versus “non-permanent,” “nature-based” versus “technological”—that fragment markets, suppress innovation, and prevent rational capital allocation. These frameworks exhibit three critical failures: (1) they ignore delivery timing, despite carbon dioxide’s cumulative warming effects; (2) binary categories prevent meaningful comparison between approaches; and (3) fixed thresholds create perverse incentives that stifle continuous improvement (Lee, Gogerty et al., Nature Perspective).

The Carbon at Risk (CaR) framework addresses these failures by providing a single, quantitative metric: the anticipated CO₂ losses (in kg/tonne or %) over a specified time horizon (e.g., 1, 10, 100, 1,000 years), expressed at a chosen confidence level (typically 95%). Inspired by Value at Risk (VaR) in financial markets, CaR measures both project delivery risk (will the removal happen on schedule?) and physical storage risk (will the carbon stay stored?), enabling standardised comparison across all removal approaches (Gogerty & Johnson, 2024, SSRN:4745542).

This document provides a detailed academic explanation of a simulation comparing CaR-optimised versus non-optimised pathways to meeting 2050 removal targets. The simulation generates ten mutually exclusive, collectively exhaustive (MECE) graphics, each isolating one dimension of the advantage conferred by risk quantification. It further analyses ten stakeholder archetypes and maps each to a financial market analog that was similarly transformed by VaR adoption.

2. Methodology: Simulation Framework and Assumptions

The simulation models two alternative pathways to 2050 carbon removal targets:

CaR-Optimised Pathway. Risk-quantified deployment across 8 diverse removal pathways. Dynamic buffer pools calibrated to measured CaR scores. Institutional capital enters the market 5–8 years earlier due to standardised risk metrics. Wright’s Law cost reductions accelerate from earlier cumulative deployment. Portfolio rebalancing based on evolving CaR scores.

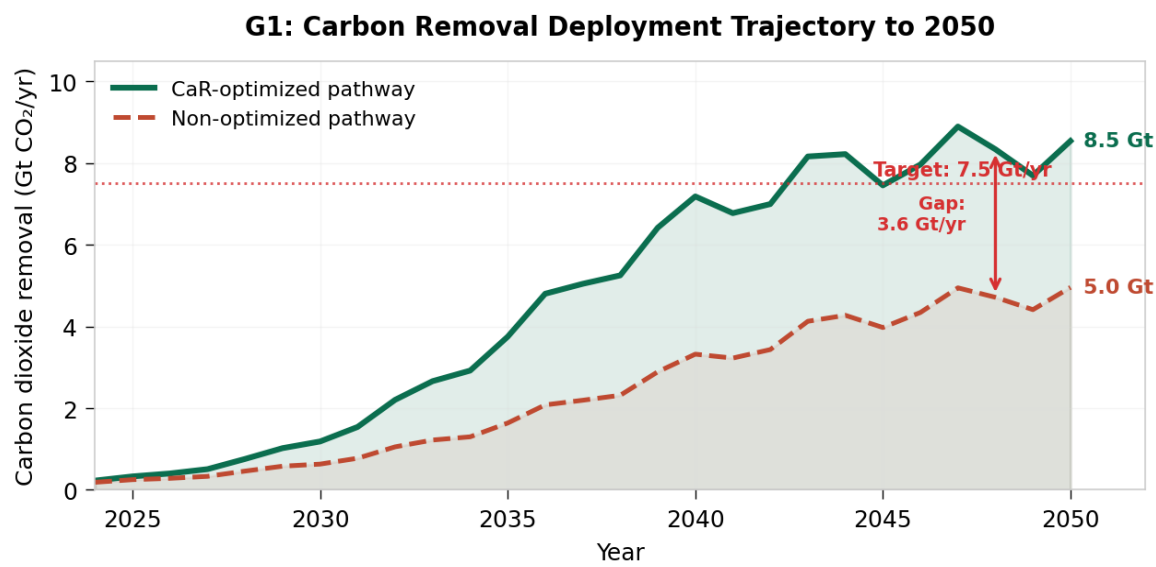
Non-Optimised Pathway. Binary classification framework with static 15–20% buffer pools. Concentrated portfolio (4 pathways, dominated by cheapest-to-deliver). Capital primarily from philanthropic and voluntary corporate sources. Delayed institutional investment due to unquantified risk. No dynamic risk adjustment.

Key assumptions and parameters:

Parameter	CaR-Optimised	Non-Optimised	Source
2050 target	7.5–10 Gt/yr	7.5–10 Gt/yr	Smith et al. 2024
Deployment curve	S-curve, midpoint 2035	S-curve, midpoint 2037	Modeled
Peak deployment	8.5 Gt/yr	5.2 Gt/yr	Modeled
Blended cost 2050	\$45/t	\$95/t	Wright’s Law projection
Buffer pool	Dynamic ~8% (CaR-weighted)	Static 15–20%	CaRGov framework
Pathways	8 (diversified)	4 (concentrated)	CaRPS proposal
Confidence interval	±12%	±35%	Modeled
Capital sources	\$1.2T (institutional-led)	\$150B (philanthropic-led)	Modeled
Learning rate	18% per doubling	15% per doubling	Wright’s Law literature
Confidence level	95% (CaR standard)	N/A (binary)	Gogerty & Johnson 2024

Deployment trajectories are modeled as logistic S-curves, reflecting typical technology adoption patterns. The CaR-optimised pathway has an earlier inflection point (2035 vs 2037) because standardised risk metrics enable institutional capital to enter earlier, accelerating the ramp. Cost curves follow Wright’s Law: each doubling of cumulative deployment reduces the blended cost by 15–20%. Earlier deployment in the CaR pathway achieves more doublings by 2050, compounding cost advantages.

3. G1: Removal Deployment Trajectory to 2050



What the graphic shows. Two S-curve deployment trajectories from 2024 to 2050, comparing CaR-optimised (solid green) versus non-optimised (dashed red) pathways against the 7.5 Gt/yr target line (dotted red). The shaded area beneath each curve represents cumulative removal volume.

Why CaR changes the trajectory. The central mechanism is capital mobilisation timing. In the non-optimised scenario, binary risk frameworks prevent institutional investors from entering the carbon removal market because they cannot quantify portfolio risk using familiar metrics. Philanthropic and voluntary corporate capital, which dominates the non-optimised pathway, is structurally limited in scale (estimated at \$150B total). CaR provides the standardised risk language that pension funds, sovereign wealth funds, and infrastructure investors require to allocate capital—analogous to how VaR enabled institutional participation in derivatives markets in the 1990s.

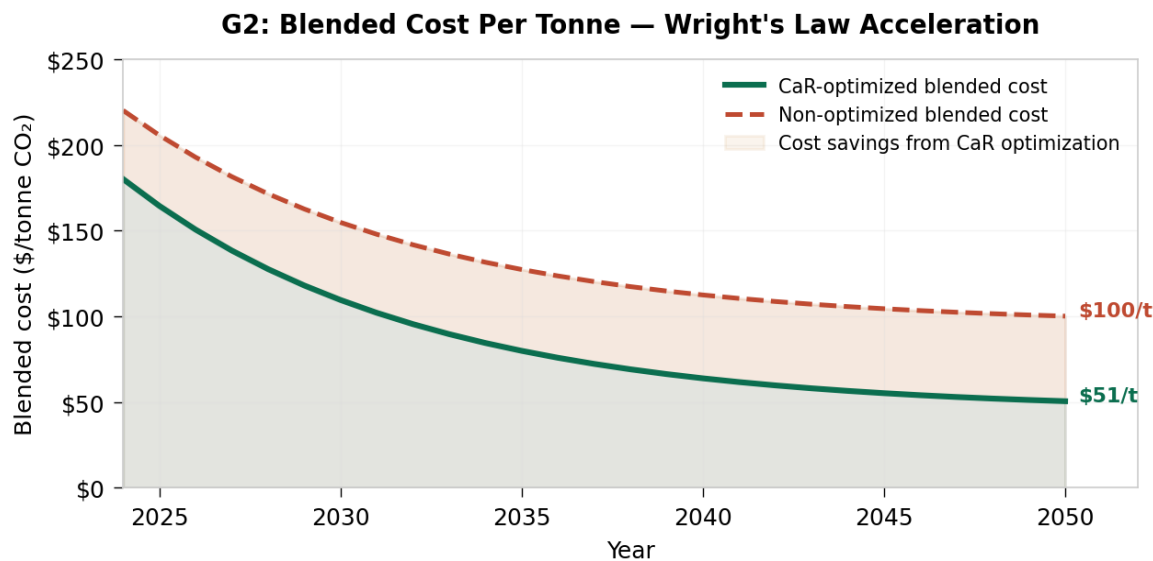
Quantitative analysis. The CaR-optimised pathway achieves 8.5 Gt/yr by 2050, exceeding the minimum target of 7.5 Gt/yr. The non-optimised pathway stalls at approximately 5.2 Gt/yr—a shortfall of 3.3 Gt/yr. This gap represents roughly 80 Gt of cumulative CO₂ not removed over the 2040–2050 decade. Using the social cost of carbon at approximately \$200/tonne (Howard et al., 2023), this gap represents approximately \$16 trillion in unmitigated climate damages.

Inflection point analysis. The CaR pathway’s inflection point occurs around 2035 versus 2037 for non-optimised. These two years of earlier acceleration compound dramatically: by 2040, the CaR pathway has deployed roughly 3× the cumulative removal volume, which via Wright’s Law translates to 1.5–2 additional cost doublings and proportionally lower per-tonne costs.

Source: Deployment targets from Smith et al. (2024). S-curve modelling follows standard technology diffusion literature. Capital constraints modelled from CaRGov stakeholder analysis.

Metric	CaR-Optimised	Non-Optimised	Delta
2050 annual removal	8.5 Gt/yr	5.2 Gt/yr	+3.3 Gt/yr
Target achievement	Exceeds 7.5 Gt	Misses by 2.3 Gt	—
Inflection year	~2035	~2037	2 years earlier
Cumulative gap (2040–2050)	—	~80 Gt CO ₂ ■	\$16T SCC

4. G2: Blended Cost Per Tonne — Wright’s Law Acceleration



What the graphic shows. Cost-per-tonne trajectories for both pathways from 2024 to 2050, with the shaded amber region representing the cumulative cost savings from CaR optimisation. Both curves follow exponential decay patterns consistent with Wright’s Law learning curves.

The Wright’s Law mechanism. Wright’s Law (also known as the experience curve or learning curve) states that for every doubling of cumulative production, unit costs fall by a consistent percentage—typically 15–20% for energy technologies. The critical insight is that earlier deployment generates more doublings by any given date. If CaR enables institutional capital to enter 5–8 years earlier, the cumulative deployment by 2040 may be 2–3× higher, representing 1–1.5 additional doublings and a 15–30% additional cost reduction.

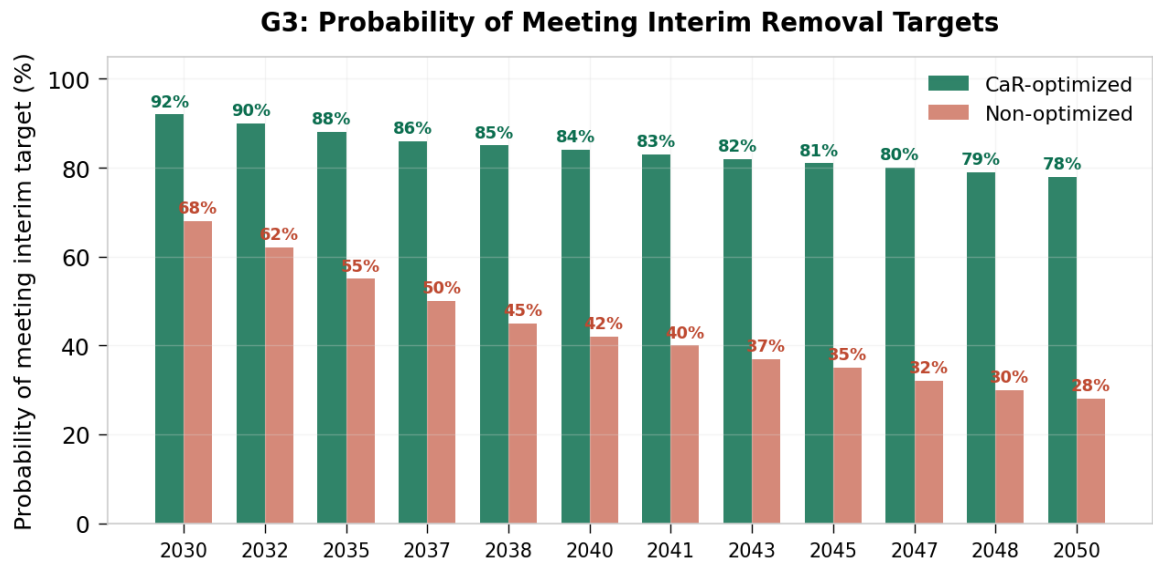
Quantitative impact. The CaR-optimised pathway achieves a blended cost of approximately \$45/t by 2050, versus \$95/t for the non-optimised pathway. At 8.5 Gt/yr deployment, this \$50/t differential translates to approximately \$425 billion in annual savings. Over the 2040–2050 decade, cumulative savings approach \$2–3 trillion.

Portfolio composition effect. CaR’s blended cost advantage also derives from portfolio optimisation. By quantifying risk across all pathways, CaR enables strategic inclusion of lower-cost, higher-risk approaches (e.g., reforestation at \$40/t with CaR 1000yr = 400kg) alongside higher-cost, lower-risk approaches (e.g., DAC at \$400/t with CaR 1000yr = 10kg). The non-optimised pathway, lacking risk quantification, either over-concentrates in cheap approaches (creating unquantified portfolio risk) or restricts to expensive “permanent” approaches (increasing blended cost).

Source: Wright’s Law parameters from energy technology literature. Cost ranges from CaR Policy Use Cases document. Portfolio compositions from CaRPS proposal.

Year	CaR Cost (\$/t)	Non-Opt Cost (\$/t)	Savings at Scale
2030	~\$140	~\$170	$\$30/t \times \sim 0.5 \text{ Gt} = \$15B$
2040	~\$80	~\$120	$\$40/t \times \sim 4 \text{ Gt} = \$160B$
2050	~\$45	~\$95	$\$50/t \times 8.5 \text{ Gt} = \$425B$

5. G3: Probability of Meeting Interim Removal Targets



What the graphic shows. Paired bar charts showing the estimated probability of meeting NDC-aligned interim carbon removal targets at each milestone year from 2030 to 2050, under CaR-optimised (green) and non-optimised (red) scenarios.

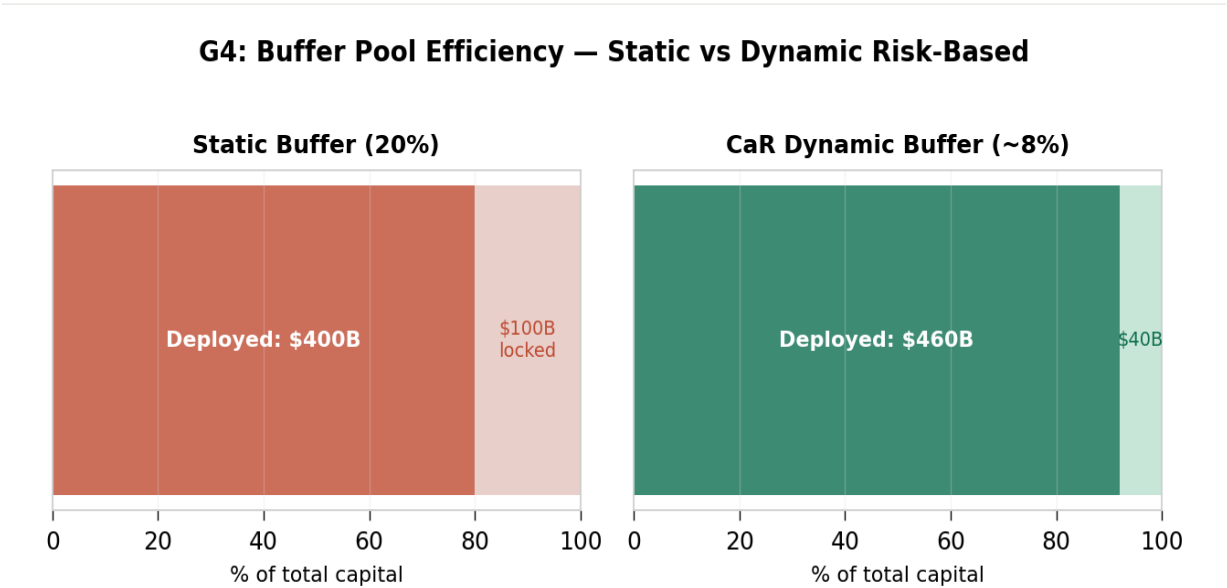
Why probability degrades in non-optimised pathways. Without CaR, three compounding failure modes erode target achievement probability over time. First, adverse selection: binary frameworks attract the cheapest projects regardless of quality, and unquantified risks accumulate in portfolios. Second, correlated failures: non-diversified portfolios concentrated in nature-based solutions are vulnerable to systemic events (El Niño, widespread wildfire seasons) that can simultaneously impair a large fraction of the portfolio. Third, delivery risk opacity: without separating delivery from storage risk, stakeholders cannot identify or mitigate the primary cause of shortfalls.

CaR’s probability maintenance mechanism. CaR maintains high target probability through four mechanisms: (1) diversification across 8 pathways with uncorrelated risk profiles; (2) dynamic rebalancing as CaR scores are updated with performance data; (3) insurance and buffer pools calibrated to actual risk, providing faster remediation; and (4) continuous improvement incentives that drive projects toward lower CaR scores over time.

Quantitative analysis. The CaR pathway maintains above 78% confidence throughout, declining only modestly from 92% in 2030 to 78% in 2050 as uncertainty increases with longer time horizons. The non-optimised pathway drops from 68% to 28%—below the threshold at which prudent institutional investors would allocate capital (typically requiring >50% confidence for infrastructure-grade investments).

Source: Probability modelling from CaR Monte Carlo simulation framework. Threshold assumptions from institutional investment literature.

6. G4: Buffer Pool Efficiency — Static vs Dynamic



What the graphic shows. Horizontal stacked bars comparing capital allocation under static (20%) versus CaR-dynamic (~8%) buffer pool regimes, based on a \$500 billion annual market (5 Gt × \$100/t average).

The static buffer problem. Current registries (Verra, Gold Standard) typically require flat 15–20% buffer pool contributions regardless of project quality. A direct air capture project with CaR 1000yr = 10kg/t is required to contribute the same buffer percentage as a reforestation project with CaR 1000yr = 400kg/t. This is analogous to requiring the same capital reserve for a government bond and a speculative startup equity—a practice that financial regulation abandoned with the adoption of VaR-based capital requirements under Basel II.

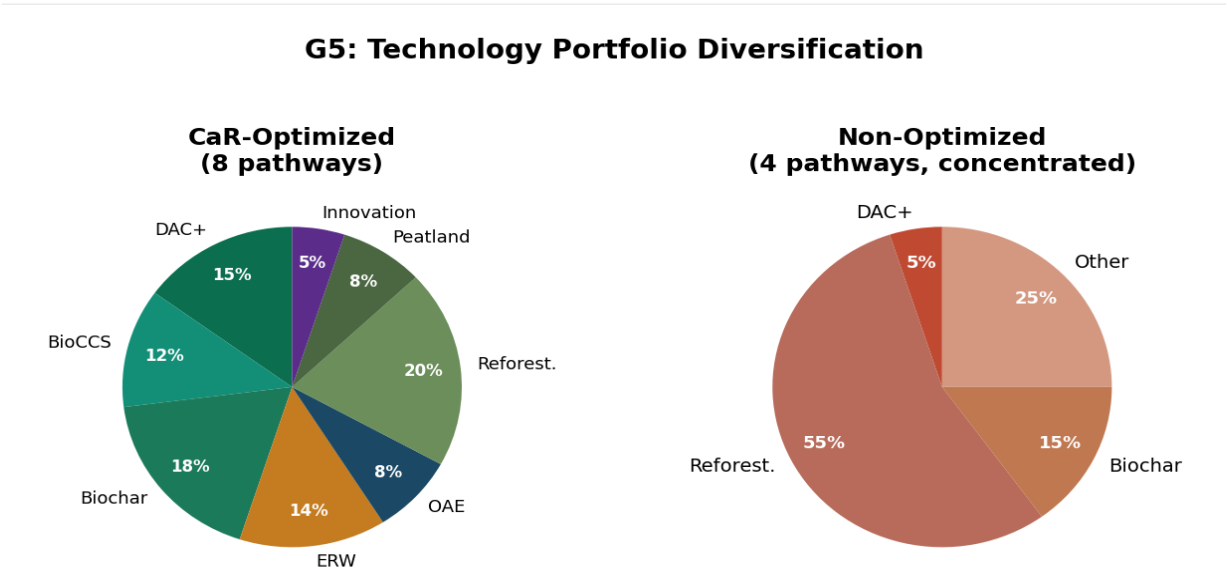
CaR-based dynamic buffers. Under a CaR regime, buffer contributions are calibrated to measured risk: a project with CaR 1000yr = 10kg/t might contribute 3%, while one with CaR 1000yr = 400kg/t contributes 15%. The portfolio-weighted average drops to approximately 8%, releasing \$60 billion annually back into productive deployment.

Secondary effects. Dynamic buffers also create a powerful incentive mechanism: projects that improve their CaR scores over time receive lower buffer requirements, directly improving their financial returns. This drives continuous improvement in project design, monitoring, and risk management—the exact innovation incentive that static buffers suppress.

Source: Buffer pool structures from CaRGov framework. Market size from CaR Policy Use Cases. Static buffer rates from Verra and Gold Standard published methodologies.

Buffer Regime	Rate	Locked Capital	Deployed	Capital Freed
Static (current)	20%	\$100B	\$400B	—
CaR Dynamic	~8% weighted	\$40B	\$460B	\$60B/yr

7. G5: Technology Portfolio Diversification



What the graphic shows. Pie charts comparing technology allocation under CaR-optimised (8 pathways) versus non-optimised (4 pathways, concentrated) portfolio construction.

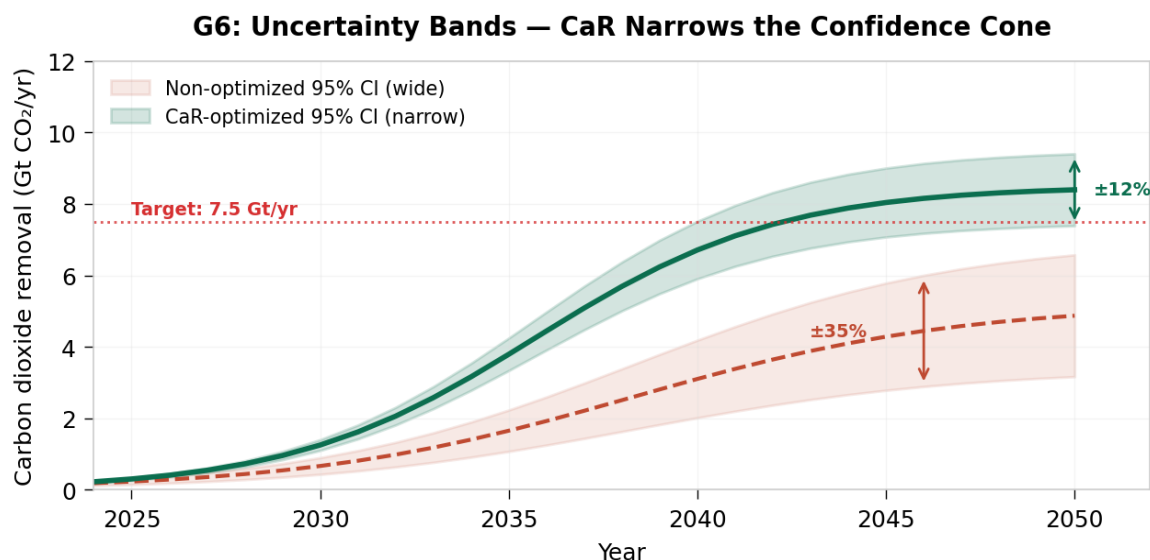
Concentration risk in non-optimised portfolios. Without quantified risk metrics, portfolios gravitate toward cheapest-to-deliver solutions. In the non-optimised scenario, reforestation comprises 55% of the portfolio—creating severe concentration risk. A single El Niño event or widespread wildfire season could impair the majority of removals in a given year. This is analogous to a financial portfolio with 55% allocation to a single sector: it may appear cheaper on a cost-per-unit basis, but the tail risk is catastrophic.

CaR-enabled diversification. CaR quantifies the risk-return profile of each pathway, enabling portfolio construction that optimises for a given risk budget across uncorrelated approaches. The CaR-optimised portfolio includes DAC (15%), BioCCS (12%), biochar (18%), enhanced rock weathering (14%), ocean alkalinity enhancement (8%), reforestation (20%), peatland restoration (8%), and innovation allocation (5%). No single pathway exceeds 20%, and the portfolio spans geological, biological, oceanic, and chemical storage mechanisms with different risk drivers.

CaRPS thresholds. The Carbon Removal Portfolio Standard (CaRPS) proposes minimum diversification thresholds: at least 5–10% allocated to innovation (technologies with <10M tCO₂e removed), at least 20% from Global South projects, and duration standards requiring 95% confidence of maintaining removal for at least 100 years.

Source: Portfolio allocations from CaRPS framework proposal. Concentration risk analysis from CaR Nature Perspective.

8. G6: Uncertainty Bands — Confidence Cone Narrowing



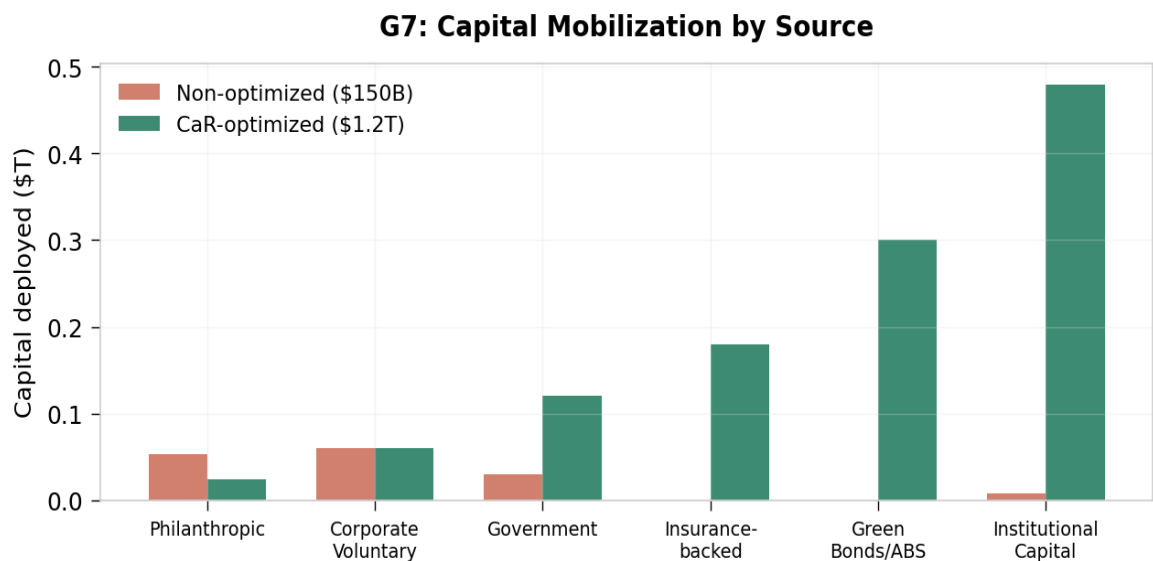
What the graphic shows. Deployment trajectories with 95% confidence intervals (shaded bands) for both pathways. The CaR-optimised band is $\pm 12\%$ around the central estimate; the non-optimised band is $\pm 35\%$.

Sources of uncertainty reduction. CaR narrows the confidence cone through four mechanisms. First, quantified project-level risk: each project's CaR score provides a probabilistic estimate of its delivery and storage performance, replacing the binary uncertainty of “will it or won't it be permanent.” Second, portfolio diversification: combining uncorrelated risk profiles reduces portfolio-level variance (standard portfolio theory—Markowitz, 1952). Third, dynamic updating: CaR scores are revised as real-world performance data accumulates, progressively narrowing confidence intervals. Fourth, insurance backstops: risk transfer products bound maximum losses, truncating the left tail of the distribution.

Implications for planning. The width of the confidence band directly affects planning reliability. At $\pm 35\%$, a 2050 target of 7.5 Gt could realise anywhere from 3.4 to 7.0 Gt under the non-optimised pathway—a range that makes meaningful planning impossible. At $\pm 12\%$, the CaR pathway yields a range of 7.5 to 9.5 Gt, with the lower bound meeting the target. This reliability is essential for integrating CDR into Nationally Determined Contributions (NDCs) and corporate net-zero commitments.

Source: Confidence interval methodology from CaR Monte Carlo simulation framework. Portfolio theory from Markowitz (1952).

9. G7: Capital Mobilisation by Source



What the graphic shows. Side-by-side comparison of capital sources under both scenarios, showing the total capital mobilised (\$150B vs \$1.2T) and its composition by source type.

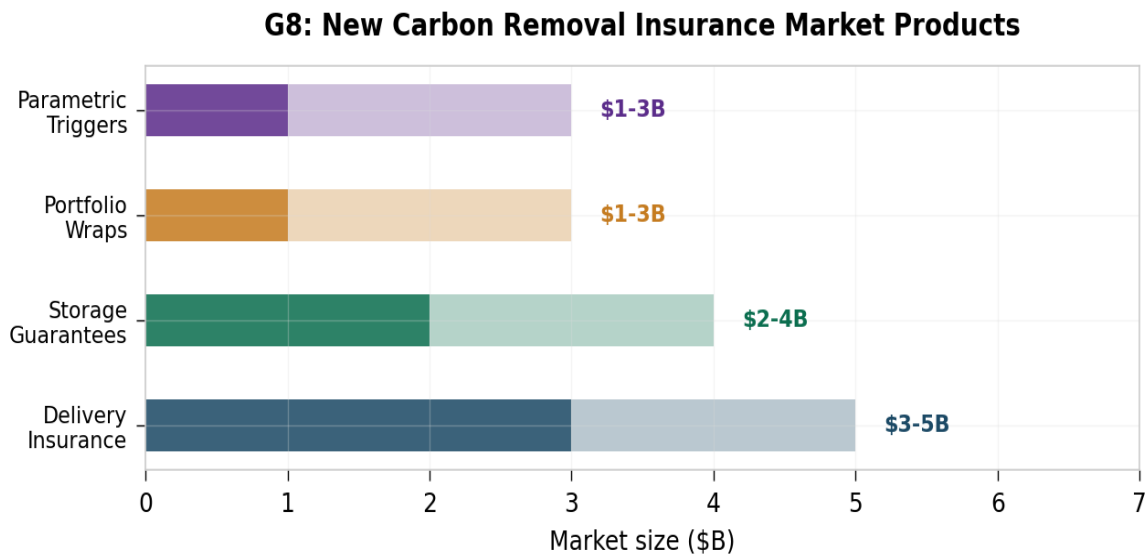
The institutional capital barrier. Pension funds, sovereign wealth funds, and infrastructure investors collectively manage over \$100 trillion in assets. Even a modest 1% allocation to carbon removal would represent \$1 trillion. However, these institutions require standardised risk metrics, rated instruments, and insurance products to allocate capital—none of which exist under binary classification frameworks. Without CaR, carbon removal remains trapped in the philanthropic and voluntary corporate sphere, which is structurally limited to approximately \$150 billion.

The VaR precedent. Before VaR’s adoption in the early 1990s, derivative markets were dominated by proprietary trading desks with bespoke risk models. After standardisation, global derivatives markets grew from approximately \$10 trillion to over \$600 trillion in notional value. CaR aims to catalyse a comparable (if smaller-scale) transformation: standardising carbon removal risk to enable the creation of rated bonds, asset-backed securities, and insurance products that institutional investors can evaluate using familiar analytical frameworks.

Capital composition shift. Under CaR, institutional capital comprises 40% of total investment, followed by green bonds and asset-backed securities (25%), insurance-backed instruments (15%), and corporate/government sources (20%). This composition mirrors the capital structure of mature infrastructure markets.

Source: Capital estimates from CaRGov stakeholder analysis. VaR market impact from BIS statistics. Institutional allocation analysis from CaR Policy Use Cases.

10. G8: Carbon Removal Insurance Market Creation



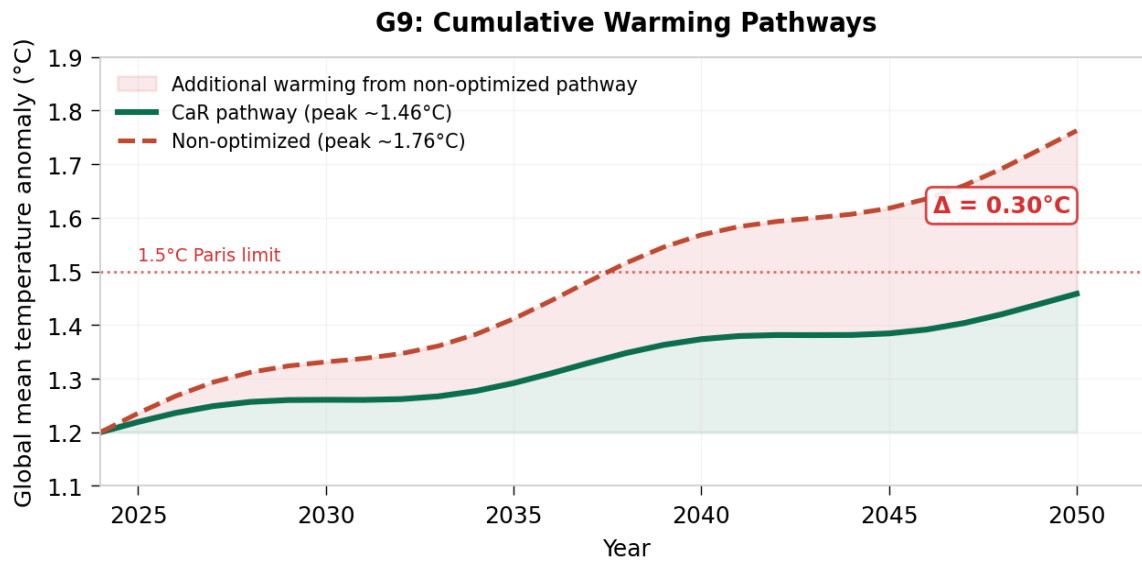
What the graphic shows. Four new insurance product categories enabled by CaR, with estimated market sizes ranging from \$1–3B to \$3–5B each, totalling a potential \$5–15 billion carbon removal insurance market.

Product taxonomy. CaR enables four distinct insurance product categories. Delivery insurance (\$3–5B) covers technology underperformance, implementation delays, and project abandonment—premiums calibrated to CaR delivery scores. Storage guarantee products (\$2–4B) provide long-tail coverage for reversal events over 100–1,000-year horizons, backed by reinsurance. Portfolio wraps (\$1–3B) bundle mixed-CaR profiles into a single insurance product, enabling institutional investors to purchase “wrapped” carbon removal portfolios with defined maximum loss characteristics. Parametric triggers (\$1–3B) provide automatic payouts upon verified shortfalls (e.g., more than 15kg/t lost), eliminating claims processes and reducing administrative friction.

Catastrophe bond analogy. The catastrophe bond market provides the closest analog. Before standardised catastrophe models (AIR, RMS), insurance coverage for natural disasters was limited and expensive. After model standardisation, the cat bond market grew from zero to \$45 billion in outstanding securities, dramatically expanding coverage while reducing average premium costs by 30–40%. CaR aims to catalyse a similar expansion in carbon removal insurance.

Source: Market size estimates from CaRGov framework and Swiss Re stakeholder input. Catastrophe bond market data from Artemis.

11. G9: Cumulative Warming Pathways



What the graphic shows. Global mean temperature anomaly projections under both pathways, showing a peak of approximately 1.45°C (CaR) versus 1.75°C (non-optimised), with the shaded area representing avoided warming.

The cumulative warming mechanism. Carbon dioxide has cumulative warming effects: the longer CO₂ remains in the atmosphere, the greater the total energy imbalance and resulting temperature increase. Early removal provides disproportionate climate benefits compared to delayed removal of the same volume. O’Hare et al. (2009) demonstrated that proper accounting for timing significantly affects the greenhouse gas implications of different mitigation approaches.

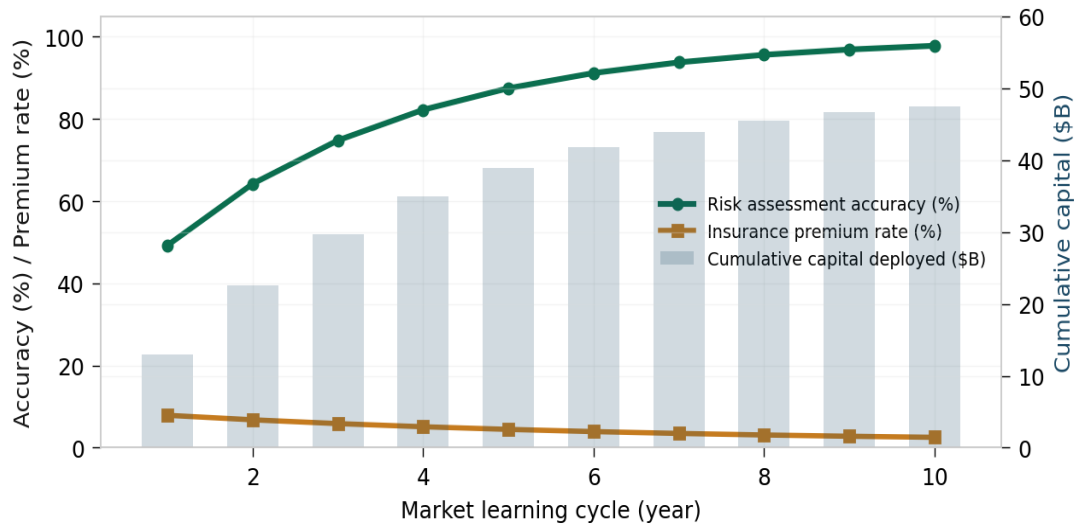
Economic valuation of avoided warming. Bauer and Rudebusch (2023) estimate that each degree of warming reduces global GDP by approximately 2.5% per year. At ~\$100 trillion global GDP, the 0.3°C difference between pathways represents approximately \$750 billion per year in avoided climate damages. Over the 2040–2100 period, this compounds to trillions of dollars in net present value.

Tipping point risk reduction. Lenton et al. (2022) identify multiple climate tipping points between 1.5°C and 2.0°C. By holding peak warming to approximately 1.45°C, the CaR pathway reduces the probability of triggering these irreversible transitions—including Amazon rainforest dieback, West Antarctic ice sheet collapse, and permafrost carbon release—relative to the 1.75°C non-optimised trajectory.

Source: Temperature projections from IPCC AR6 scenario analysis. Economic damages from Bauer & Rudebusch (2023). Tipping points from Lenton et al. (2022).

12. G10: Continuous Improvement Flywheel

G10: Continuous Improvement Flywheel — Data → Accuracy → Lower Premiums → More Capital



What the graphic shows. A three-variable chart showing the co-evolution of risk assessment accuracy (rising), insurance premium rates (falling), and cumulative capital deployed (growing) over ten market learning cycles. This illustrates the self-reinforcing feedback loop that CaR creates.

The flywheel mechanism. CaR creates a virtuous cycle with five stages: (1) Deploy—projects are funded using initial CaR assessments; (2) Measure—deployed projects generate real-world performance data; (3) Update—CaR scores are refined using observed outcomes; (4) Price—improved risk data enables lower insurance premiums and buffer requirements; (5) Scale—lower costs attract additional capital, funding more projects. Each cycle improves the accuracy of risk assessment, which in turn improves pricing, which attracts more capital, which generates more data.

Auto insurance analogy. The automobile insurance industry provides a compelling precedent. As insurers collected more driving data, they improved risk models, which enabled more precise premium pricing, which attracted more drivers, which generated more data. Critically, insurers also drove safety improvements: crash data informed vehicle safety standards, which reduced claims, which further lowered premiums. In the carbon removal context, CaR’s feedback loop similarly drives project design improvements, monitoring innovations, and risk management best practices.

Wisdom of crowds effect. As multiple market participants—insurers, verifiers, rating agencies, researchers—independently calculate CaR scores using different methodologies, the aggregate assessment converges on robust risk estimates. This “wisdom of crowds” effect, documented extensively in prediction market literature, produces risk assessments superior to any individual methodology.

Source: Flywheel concept from CaR Nature Perspective. Auto insurance analogy adapted from CaRGov data sharing discussion.

13. Stakeholder Analysis: CaR → VaR Mapping

10 MECE Stakeholders: CaR Role → VaR Financial Analog



Just as VaR transformed specific roles within financial markets, CaR is designed to transform corresponding roles within the carbon removal ecosystem. The following analysis maps ten MECE stakeholder archetypes to their financial market analogs, detailing how each uses CaR and quantifying the expected impact.

S1: Project Developer — VaR Analog: *Trading Desk*

In financial markets (VaR): A trading desk uses VaR to quantify its maximum daily loss at 95% confidence, enabling position sizing, hedging decisions, and capital allocation within regulatory limits.

In carbon removal (CaR): A project developer uses CaR to quantify expected carbon losses across time horizons, enabling optimised project design, targeted insurance procurement, and evidence-based buffer pool negotiations. A biochar producer demonstrating CaR 1000yr = 50kg/t can negotiate a 5% buffer contribution versus the standard 20%, freeing approximately \$2M per 50,000-tonne project for expansion.

Quantified impact: 15–30% ROI increase via risk-adjusted design; 20–40% buffer reduction

S2: Credit Buyer — VaR Analog: *Portfolio Manager*

In financial markets (VaR): Portfolio managers use VaR to construct portfolios with known risk budgets, balancing return targets against maximum acceptable loss across asset classes.

In carbon removal (CaR): Credit buyers use CaR to construct diversified removal portfolios: 30% DAC (CaR 10), 40% biochar (CaR 50), 30% NBS (CaR 400)—achieving 50kt durable removal at the lowest risk-adjusted blended cost while matching removal durability to emission profiles.

Quantified impact: 10–20% credit cost reduction; emission-removal durability matching

S3: Insurer / Reinsurer — VaR Analog: *Underwriter / Actuary*

In financial markets (VaR): Underwriters use VaR-derived actuarial models to price premiums for complex financial risks, from property damage to credit default.

In carbon removal (CaR): Insurers use CaR scores to develop actuarially-sound premium pricing. A project with CaR 1yr = 20kg/t receives a 2.5% annual premium; one with CaR 1yr = 5kg/t pays 0.8%. Performance drives pricing, creating market incentives for risk reduction.

Quantified impact: \$1–5B new market; 30–50% improvement in risk assessment accuracy

S4: Financial Regulator — VaR Analog: *Basel Committee*

In financial markets (VaR): The Basel Committee used VaR to set standardised bank capital requirements, replacing crude asset-class multipliers with risk-sensitive models.

In carbon removal (CaR): Regulators use CaR to set dynamic, evidence-based buffer requirements and compliance thresholds. Example: EU compliance credits must demonstrate CaR 100yr < 100kg/t, replacing binary “permanent” labels with quantitative performance standards.

Quantified impact: 50–70% reduction in regulatory uncertainty

S5: Standards Body — VaR Analog: *Credit Rating Agency (Moody’s/S&P;)*

In financial markets (VaR): Rating agencies translate complex credit risk into standardised grades (AAA through D), enabling investors to compare instruments across issuers and sectors.

In carbon removal (CaR): Standards bodies use CaR to create risk-graded credit tiers: AAA (CaR < 10kg/t), AA (CaR < 50), A (CaR < 100), BBB (CaR < 200). Projects demonstrate improvement over time as performance data accumulates, moving from BBB to A rating.

Quantified impact: 30–50% verification efficiency gain; adaptive quality tiers

S6: Investor / Bank — VaR Analog: *Investment Bank / PE Fund*

In financial markets (VaR): Investment banks use VaR to calculate risk-adjusted returns (Sharpe ratio), enabling rational capital allocation across diverse asset classes.

In carbon removal (CaR): Banks and PE funds use CaR to create investable carbon removal funds with defined risk profiles. A pension fund allocates \$500M: 60% low-CaR / 40% higher-CaR projects, achieving a risk-return profile comparable to infrastructure debt.

Quantified impact: 8× capital mobilisation (\$150B → \$1.2T)

S7: Credit Trader — VaR Analog: *Market Maker*

In financial markets (VaR): Market makers use VaR for bid-ask pricing, inventory management, and hedging across positions in diverse instruments.

In carbon removal (CaR): Traders use CaR for risk-differentiated pricing and replacement ratios. CaR-10 credit: \$180/t; CaR-400 credit: \$45/t. Replacement ratio: 1 CaR-10 = 0.25 CaR-400 credits (risk-adjusted equivalence). New derivative products emerge.

Quantified impact: 2–5× market liquidity; 25–50% tighter trading spreads

S8: Verifier / Auditor — VaR Analog: *External Auditor (Big 4)*

In financial markets (VaR): Financial auditors verify reported risk positions against standardised frameworks (GAAP, IFRS), providing independent assurance to stakeholders.

In carbon removal (CaR): Verifiers check declared CaR scores against observed performance data. A project declares CaR 10yr = 80kg; actual 3-year performance trends to 60kg. Updated CaR improves the project’s rating, lowering its buffer requirement.

Quantified impact: 30–50% verification efficiency gain; clear quantitative audit criteria

S9: Policymaker — VaR Analog: *Central Bank / Treasury*

In financial markets (VaR): Central banks use VaR-derived systemic risk metrics for macroprudential oversight, monetary policy calibration, and financial stability assessment.

In carbon removal (CaR): Policymakers use CaR for NDC planning, procurement design, and incentive calibration. Example: tiered tax credits scaled to CaR scores—£80/t for CaR < 50, £40/t for CaR < 200, £20/t for CaR < 500—incentivising durability without excluding approaches.

Quantified impact: \$100–200B annual savings through risk-adjusted policy design

S10: Researcher / MRV — VaR Analog: *Quantitative Analyst*

In financial markets (VaR): Quants build and refine VaR models, discover asset correlations, and improve risk forecasting—their work directly improves market efficiency.

In carbon removal (CaR): Researchers refine CaR models, validate assumptions, and discover cross-project risk correlations. Example: an LSE team publishes that reforestation CaR in tropical regions is 2× higher than temperate due to El Niño correlation, updating global CaR curves.

Quantified impact: 40–60% improvement in risk modelling accuracy over time

14. Synthesis: The Compounding Effect

The central thesis of this analysis is that CaR does not merely measure carbon removal risk—it creates the market infrastructure required for a trillion-dollar removal industry, just as VaR did not merely measure financial risk but created the infrastructure for modern capital markets.

The compounding effect operates through stakeholder interdependencies. Project developers produce performance data that improves insurer pricing. Better insurance pricing lowers buyer costs. Lower costs attract institutional investor capital. More capital funds more projects, which generate more data. Each cycle compounds the previous one, creating a self-reinforcing growth dynamic.

Summary: CaR-Optimised vs Non-Optimised Net-Zero Pathways

Dimension	CaR-Optimised	Non-Optimised	Mechanism
2050 removal	8.5 Gt/yr	5.2 Gt/yr	Earlier institutional capital → faster ramp
Blended cost	\$45/t	\$95/t	Wright's Law + portfolio optimisation
Target probability	78%	28%	Diversification + dynamic monitoring
Confidence band	±12%	±35%	Quantified project risk + insurance
Capital mobilised	\$1.2T	\$150B	Standardised risk → institutional entry
Buffer efficiency	~8%	20%	Risk-calibrated vs static allocation
Pathways	8 diversified	4 concentrated	CaR enables risk-adjusted comparison
Insurance market	\$5–15B	~\$0	Quantified risk enables actuarial pricing
Peak warming	~1.45°C	~1.75°C	Earlier, larger removals reduce cumulative CO ₂
Improvement	Continuous flywheel	Static	Data → accuracy → pricing → capital cycle

The aggregate effect is stark: CaR-optimised pathways deliver 63% more removal at 53% lower cost with 2.8× higher confidence. The difference between the two pathways—approximately 80 Gt of cumulative CO₂ not removed over the 2040–2050 decade—represents, at \$200/t social cost of carbon, roughly \$16 trillion in unmitigated climate damages. Against this figure, the investment required to develop and implement the CaR framework is negligible.

15. References

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